NutriMadhvi — Architecture

Native Android · Kotlin + Jetpack Compose · Room v30 · Gemini multimodal · ~45k LOC / 119 files

UI LAYER · JETPACK COMPOSE

ChatScreen
meal photo, streaming replies, action chips

Dashboard / Análise
macro bars, body-comp multi-line charts

FoodHabits · History
live search, edit-in-place, action log

Voice Mode
🔊 TtsController · 🎙 SpeechRecognizer

ChatViewModel holds UI state → dispatches intents downward, renders streamed tokens & proposals upward

user query · meal image ↓ ↑ streamed tokens · proposals

CHAT ORCHESTRATION · LOCAL-FIRST

IntentRouter
regex match — resolve facts locally before any LLM call

DataFetcher
parallel snapshot gather (meals · activity · body-comp)

ChatOrchestrator
LLM only for synthesis · ModelCircuitBreaker · PredictionEngine

Local-first: route → fetch in parallel → LLM synthesises. Facts are measured, not guessed.

→ image + prompt

MEAL ANALYSIS · VISION CASCADE

MealAnalysisOrchestrator
tiered vision-safe cascade, streaming events

PartialJsonExtractor
salvage macros from truncated stream

Cluster Enrichment
macros from user's own history (median /100g)

3.6-flash → escalate 3.5-flash if low confidence → text fallback

build request payload ↓ ↑ parsed proposals / macros

GEMINI ENGINE · REMOTE

GeminiRepository
model selection · fallback

GeminiStreamingClient
SSE · finishReason==STOP gate

GeminiCacheClient
stable-prefix prompt caching

ContextBuilder
body-comp series · rolling window

ActionProposalParser
LOG_MEAL / ACTIVITY / CHART tags

HallucinationDetector
habit guard · stream watchdog

→ read / write snapshots

PERSISTENCE · ROOM V30

FoodDao
entries · ingredients · editLog

ChatDao
messages · FTS4 search

BodyCompositionDao
weight/muscle/fat/water %

Bmr / Target Snapshots
versioned, append-only

DailyActivityDao
SH daily cache

PredictedQa · Insights
warm answers · KB

30 migrations · WAL · food_images on filesDir · never destructive on user data

HTTPS · x-goog-api-key header ↓ ↑ SDK reads · SAF writes

EXTERNAL · AI

Google Gemini API
3.6/3.5-flash vision · flash-lite · Tier 1 quotas

EXTERNAL · HEALTH & SENSORS

Samsung Health Data SDK
active kcal · workouts · weight · body composition (AAR 1.1.0)

EXTERNAL · EXPORT & BG

Drive (SAF) · IPMA · WorkManager
XLSX/HTML report · weather · daily snapshot & backup workers